

# Amazon Procurement Workbook

Prices and ASINs marked unverified need same-day procurement verification. Last checked 2026-05-27.

## DASHBOARD

|                                      |                                   |                                     |                                     |
|--------------------------------------|-----------------------------------|-------------------------------------|-------------------------------------|
| <b>\$1,357.47</b><br>PREFERRED TOTAL | <b>\$890.78</b><br>LOW-COST TOTAL | <b>\$438</b><br>FROM TREES REQUIRED | <b>\$177.98</b><br>PY-STEM REQUIRED |
| <b>\$104</b><br>SAFETY/PPE REQUIRED  | <b>\$107</b><br>PRIZE TOTAL       | <b>6</b><br>UNVERIFIED LINE COUNT   | <b>0</b><br>SEARCH-ONLY LINE COUNT  |

## SPEND BY PRIORITY

| PRIORITY    | ESTIMATED SUBTOTAL |
|-------------|--------------------|
| required    | \$996.98           |
| recommended | \$332.49           |
| prize       | \$107              |
| backup-only | \$96               |

## SPEND BY CAMP

| CAMP               | ESTIMATED SUBTOTAL |
|--------------------|--------------------|
| From Trees to Tech | \$604              |
| PY-STEM            | \$311.47           |
| Shared             | \$414              |

**Safety goggles, class set**

REQUIRED

\$30 UNIT

\$30 TOTAL

LOCAL ESTIMATE

SHARED · ALL · ALL ACTIVITIES · PPE · REUSABLE · CHECKED 2026-05-27

NEED 30 goggles · SPARE 0 if existing · BUY 1 · ON HAND maybe yes · LOCAL BETTER yes

PICK classroom safety goggles 30 pack · FIND Amazon search: classroom safety goggles 30 pack · CHEAPER Borrow from science stockroom

PURPOSE Eye protection for splash, soil, plant peel, and oobleck activities

SAFETY Sanitize between users.

STORAGE PPE bin

SOURCE ACS/NSTA general lab safety

NOTES Safety-critical.

**Nitrile gloves, 100 pack**

REQUIRED

\$12 UNIT

\$24 TOTAL

LOCAL ESTIMATE

SHARED · ALL · ALL ACTIVITIES · PPE · CONSUMABLE · CHECKED 2026-05-27

NEED 2 boxes · SPARE 15 percent built into second box · BUY 2 · ON HAND maybe yes · LOCAL BETTER yes

PICK nitrile gloves medium 100 pack · FIND Amazon search: nitrile gloves medium 100 pack · CHEAPER School stockroom

PURPOSE Soil, plant, oobleck, and cleanup handling

SAFETY Latex-free preferred.

STORAGE PPE bin

SOURCE ACS/NSTA general lab safety

NOTES Safety-critical.

**Clipboards**

RECOMMENDED

\$18 UNIT

\$18 TOTAL

LOCAL ESTIMATE

SHARED · ALL · ALL ACTIVITIES · MEASUREMENT · REUSABLE · CHECKED 2026-05-27

NEED 30 · SPARE 0 if existing · BUY 1 · ON HAND maybe yes · LOCAL BETTER yes

PICK clipboards classroom pack · FIND Amazon search: clipboards classroom pack · CHEAPER Borrow from department

PURPOSE Outdoor and room data collection

STORAGE Field bin

SOURCE planning line

NOTES Reusable.

**Stopwatches or phone timers**

RECOMMENDED

\$15 UNIT

\$15 TOTAL

LOCAL ESTIMATE

SHARED · ALL · ALL ACTIVITIES · MEASUREMENT · REUSABLE · CHECKED 2026-05-27

NEED 6 to 12 · SPARE 0 if phones allowed · BUY 1 · ON HAND maybe yes · LOCAL BETTER yes

PICK digital stopwatch 12 pack · FIND Amazon search: digital stopwatch 12 pack · CHEAPER Use projected timer plus staff phones

PURPOSE Timing races, trials, and relays

STORAGE Measurement bin

SOURCE planning line

NOTES Reusable.

## Masking tape and painter tape

REQUIRED

\$18 UNIT

\$18 TOTAL

LOCAL ESTIMATE

SHARED · ALL · ALL ACTIVITIES · SUPPLIES · CONSUMABLE · CHECKED 2026-05-27

NEED 6 rolls · SPARE 1 spare roll · BUY 1 · ON HAND maybe yes · LOCAL BETTER yes

PICK masking tape multi pack · FIND Amazon search: masking tape multi pack · CHEAPER School supply closet

PURPOSE Lane marking, labels, and safe floor routes

SAFETY Check floor compatibility before use.

STORAGE Tape bin

SOURCE planning line

NOTES Avoid residue.

## Cardstock, copy paper, markers

REQUIRED

\$35 UNIT

\$70 TOTAL

LOCAL ESTIMATE

SHARED · ALL · ALL ACTIVITIES · SUPPLIES · CONSUMABLE · CHECKED 2026-05-27

NEED 500 to 800 pages plus markers · SPARE 15 percent print buffer · BUY 2 · ON HAND maybe yes · LOCAL BETTER yes

PICK cardstock 250 sheets and markers · FIND Amazon search: cardstock 250 pack markers · CHEAPER Local print shop or school supply

PURPOSE Cards, score slips, station signs, and builds

STORAGE Print bin

SOURCE planning line

NOTES Printing lines separated in Printing sheet.

**MudWatt or equivalent microbial fuel cell kits**

REQUIRED

\$45 UNIT

\$270 TOTAL

UNVERIFIED

FROM TREES TO TECH · TTT-01 · MUDWATT BIOELECTRIC LEAGUE · SPECIALTY KIT · REUSABLE PLUS CONSUMABLE  
SOIL · CHECKED 2026-05-27

NEED 6 kits preferred · SPARE 1 spare cell if possible · BUY 6 · ON HAND unknown · LOCAL BETTER no

PICK MudWatt Grow Your Living Fuel Cell search · FIND Amazon search: MudWatt STEM kit · CHEAPER Borrow existing kits or purchase direct from Magical Microbes/Home Science Tools

PURPOSE Weeklong soil electricity anchor

SAFETY Gloves and handwashing. No food near mud cells.

STORAGE MudWatt bin, store dry between uses

SOURCE <https://www.magicalmicrobes.com/pages/how-mudwatt-works>

NOTES Main procurement risk.

**Fresh soil or mud samples**

REQUIRED

LOCAL ESTIMATE

FROM TREES TO TECH · TTT-01 · MUDWATT BIOELECTRIC LEAGUE · CONSUMABLE · CONSUMABLE · CHECKED 2026-05-27

NEED 6 sealed containers · SPARE 15 percent extra · BUY 1 · ON HAND no · LOCAL BETTER yes

PICK local soil or potting soil · FIND local purchase or collection · CHEAPER Collect from approved campus site

PURPOSE Microbial fuel source

SAFETY Avoid contaminated soil. Gloves required.

STORAGE Sealed soil tub

SOURCE <https://www.magicalmicrobes.com/pages/how-mudwatt-works>

NOTES Use approved campus collection only.

**Mini hygrometer/thermometer units**

RECOMMENDED

\$12 UNIT

\$24 TOTAL

LOCAL ESTIMATE

FROM TREES TO TECH · TTT-02 · FOREST SENSOR SPRINT · FIELD MEASUREMENT · REUSABLE · CHECKED 2026-05-27

NEED 6 to 8 units · SPARE 1 spare · BUY 2 · ON HAND no · LOCAL BETTER yes

PICK mini digital hygrometer thermometer · FIND Amazon search: mini digital hygrometer thermometer · CHEAPER Borrow from lab

PURPOSE Microclimate data collection

STORAGE Sensor bin

SOURCE <https://observer.globe.gov/do-globe-observer/trees>

NOTES Use relative comparisons if readings drift.

**3-in-1 soil moisture meter**

RECOMMENDED

\$12 UNIT

\$72 TOTAL

UNVERIFIED

FROM TREES TO TECH · TTT-02 · FOREST SENSOR SPRINT · FIELD MEASUREMENT · REUSABLE · CHECKED 2026-05-27

NEED 6 units · SPARE 0 to 1 spare · BUY 6 · ON HAND no · LOCAL BETTER yes

PICK SONKIR Soil pH Meter MS02 3-in-1 · FIND [amazon.com/dp/B07BR52P26](https://amazon.com/dp/B07BR52P26) · ASIN B07BR52P26 · CHEAPER Moisture-only meter

PURPOSE Optional soil comparison

SAFETY Probe tips are pointed.

STORAGE Wipe probes clean

SOURCE <https://observer.globe.gov/do-globe-observer/trees>

NOTES Price hidden or not confirmed.

## Paper clinometer templates

REQUIRED

LOCAL ESTIMATE

FROM TREES TO TECH · TTT-02 · FOREST SENSOR SPRINT · PRINTING · CONSUMABLE · CHECKED 2026-05-27

NEED 30 sheets · SPARE 5 spare · BUY 1 · ON HAND no · LOCAL BETTER yes

PICK local print · FIND local purchase · CHEAPER Hand-draw masters

PURPOSE Tree height estimates

STORAGE Field folder

SOURCE <https://observer.globe.gov/do-globe-observer/trees>

NOTES Use GLOBE-style clinometer.

## Paper, cardstock, paper clips, string

REQUIRED

\$12 UNIT

\$12 TOTAL

LOCAL ESTIMATE

FROM TREES TO TECH · TTT-03 · SEED DISPERSAL DERBY · BUILD SUPPLIES · CONSUMABLE · CHECKED 2026-05-27

NEED 6 team kits · SPARE 15 percent extra · BUY 1 · ON HAND maybe yes · LOCAL BETTER yes

PICK cardstock paper clips string bundle · FIND Amazon search: cardstock paper clips string · CHEAPER Use existing office supplies

PURPOSE Passive seed prototypes

SAFETY No throwing at people. Low-power fan only.

STORAGE Seed bin

SOURCE <https://www.sciencebuddies.org/stem-activities/gone-with-the-wind-plant-seed-dispersal>

NOTES No paper airplane or rocket builds.

## Paper towels, cotton string, cups, trays

REQUIRED

\$18 UNIT

\$18 TOTAL

LOCAL ESTIMATE

FROM TREES TO TECH · TTT-04 · XYLEM PIPELINE RELAY · LAB SUPPLIES · CONSUMABLE · CHECKED 2026-05-27

NEED 6 tray stations · SPARE 15 percent extra · BUY 1 · ON HAND maybe yes · LOCAL BETTER yes

PICK paper towels cotton string plastic cups trays · FIND Amazon search: paper towels cotton string plastic cups trays · CHEAPER Dollar store or school stock

PURPOSE Capillary transport race

SAFETY Use trays. Wipe spills immediately.

STORAGE Water activity bin

SOURCE <https://www.sciencebuddies.org/stem-activities/gone-with-the-wind-plant-seed-dispersal>

NOTES Use washable dye only.

## Plant profile cards and control dial boards

REQUIRED

\$15 UNIT

\$15 TOTAL

LOCAL ESTIMATE

FROM TREES TO TECH · TTT-05 · GREENHOUSE CLIMATE CONTROLLER CHALLENGE · PRINTING · REUSABLE · CHECKED 2026-05-27

NEED 6 laminated sets · SPARE 1 spare set · BUY 1 · ON HAND no · LOCAL BETTER yes

PICK local print and laminate · FIND local purchase · CHEAPER Sheet protectors instead of lamination

PURPOSE Greenhouse settings game

SAFETY Stay on tour route. Do not touch restricted controls.

STORAGE Greenhouse folder

SOURCE <https://ambler.temple.edu/campus-resources/greenhouse>

NOTES Use actual tour evidence.

## Dry pine cones

RECOMMENDED

LOCAL ESTIMATE

FROM TREES TO TECH · TTT-06 · PINECONE WEATHER MACHINE CHALLENGE · NATURAL MATERIAL · REUSABLE · CHECKED 2026-05-27

**NEED** 12 to 18 cones · **SPARE** 3 spare · **BUY** 1 · **ON HAND** maybe no · **LOCAL BETTER** yes

**PICK** local nature collection · **FIND** local collection · **CHEAPER** Ask grounds crew

**PURPOSE** Humidity-responsive device

**SAFETY** Inspect for insects, sap, and sharp edges.

**STORAGE** Dry natural-materials bin

**SOURCE** <https://asknature.org/strategy/pine-cones-open-and-close-in-response-to-weather/>

**NOTES** Dry before use.

## Spray bottles, index cards, skewers, rulers

REQUIRED

\$20 UNIT

\$20 TOTAL

LOCAL ESTIMATE

FROM TREES TO TECH · TTT-06 · PINECONE WEATHER MACHINE CHALLENGE · SUPPLIES · MIXED · CHECKED 2026-05-27

**NEED** 6 stations · **SPARE** 15 percent extra · **BUY** 1 · **ON HAND** maybe yes · **LOCAL BETTER** yes

**PICK** spray bottles index cards skewers rulers · **FIND** Amazon search: spray bottles index cards skewers rulers · **CHEAPER** Use existing classroom supplies

**PURPOSE** Weather machine build

**SAFETY** Keep floors dry after spray.

**STORAGE** Weather machine bin

**SOURCE** <https://asknature.org/strategy/pine-cones-open-and-close-in-response-to-weather/>

**NOTES** Table-based.

## Plant and pollinator card decks

REQUIRED

\$25 UNIT

\$25 TOTAL

LOCAL ESTIMATE

FROM TREES TO TECH · TTT-07 · POLLINATOR NETWORK DRAFT AND BUILD · PRINTING · REUSABLE · CHECKED 2026-05-27

**NEED** 6 decks · **SPARE** 1 spare deck · **BUY** 1 · **ON HAND** no · **LOCAL BETTER** yes

**PICK** local print and laminate · **FIND** local purchase · **CHEAPER** Sheet protectors

**PURPOSE** Pollinator habitat strategy game

**SAFETY** No live insects.

**STORAGE** Pollinator folder

**SOURCE** <https://www.usda.gov/farming-and-ranching/plants-and-crops/plant-breeding/pollinators>

**NOTES** Print in color if possible.

## Clue cards, maps, evidence tokens

REQUIRED

\$20 UNIT

\$20 TOTAL

LOCAL ESTIMATE

FROM TREES TO TECH · TTT-08 · ARBORETUM ECO-QUEST · PRINTING · REUSABLE · CHECKED 2026-05-27

**NEED** 6 route packs · **SPARE** 1 spare pack · **BUY** 1 · **ON HAND** no · **LOCAL BETTER** yes

**PICK** local print and laminate · **FIND** local purchase · **CHEAPER** Black and white route cards

**PURPOSE** Outdoor checkpoint challenge

**SAFETY** Buddy system. Approved route only.

**STORAGE** Arboretum folder

**SOURCE** <https://observer.globe.gov/do-globe-observer/trees>

**NOTES** Confirm route with host.

## Minecraft Education devices and world template

REQUIRED

FROM TREES TO TECH · TTT-09 · MINECRAFT TREE WORLD RESILIENCE CUP · SOFTWARE · REUSABLE · CHECKED 2026-05-27

NEED 6 devices plus projector · SPARE 0 · BUY 0 · ON HAND unknown · LOCAL BETTER yes

PICK Minecraft Education licenses · FIND education.minecraft.net/en-us · CHEAPER Paper resilience design board

PURPOSE Digital ecosystem build

SAFETY Standard device use and login management.

STORAGE Digital kit

SOURCE <https://education.minecraft.net/en-us/resources/science/biodiversity>

NOTES Confirm licensing.

## Tree ring image cards and answer boards

REQUIRED

\$20 UNIT

\$20 TOTAL

LOCAL ESTIMATE

FROM TREES TO TECH · TTT-10 · TREE RING CLIMATE DETECTIVE TOURNAMENT · PRINTING · REUSABLE · CHECKED 2026-05-27

NEED 6 sets · SPARE 1 spare · BUY 1 · ON HAND no · LOCAL BETTER yes

PICK local print and laminate · FIND local purchase · CHEAPER Wood cookies if already owned

PURPOSE Climate proxy data tournament

SAFETY Inspect wood slices for splinters if used.

STORAGE Tree ring folder

SOURCE <https://www.ncei.noaa.gov/products/paleoclimatology/tree-ring>

NOTES Use printed images for safer reset.

## Pipettes, wax paper, sandpaper, ramps

REQUIRED

\$18 UNIT

\$18 TOTAL

LOCAL ESTIMATE

FROM TREES TO TECH · TTT-11 · LOTUS LEAF SURFACE SPRINT · LAB SUPPLIES · MIXED · CHECKED 2026-05-27

NEED 6 team sets · SPARE 15 percent extra · BUY 1 · ON HAND maybe no · LOCAL BETTER yes

PICK transfer pipettes wax paper sandpaper · FIND Amazon search: transfer pipettes wax paper sandpaper · CHEAPER Science stockroom

PURPOSE Hydrophobic surface tests

SAFETY Keep floors dry. Washable dirt proxy only.

STORAGE Surface sprint bin

SOURCE <https://asknature.org/strategy/surface-allows-self-cleaning/>

NOTES No major staining.

## Clear nail polish, tape, leaf samples

REQUIRED

\$20 UNIT

\$20 TOTAL

LOCAL ESTIMATE

FROM TREES TO TECH · TTT-12 · LEAF STOMATA MICROSCOPE DETECTIVE · LAB SUPPLIES · CONSUMABLE · CHECKED 2026-05-27

NEED 6 team sets · SPARE 15 percent extra · BUY 1 · ON HAND no · LOCAL BETTER yes

PICK clear nail polish tape microscope slides · FIND Amazon search: clear nail polish tape microscope slides · CHEAPER Use phone macro lens and prepared peels

PURPOSE Leaf peel microscopy detective activity

SAFETY Ventilation for nail polish. Gloves optional.

STORAGE Stomata bin

SOURCE <https://observer.globe.gov/do-globe-observer/trees>

NOTES Legacy-safe replacement.

USB digital microscope or classroom  
microscopes

RECOMMENDED

\$35 UNIT

\$70 TOTAL

LOCAL ESTIMATE

FROM TREES TO TECH · TTT-12 · LEAF STOMATA MICROSCOPE DETECTIVE · EQUIPMENT · REUSABLE · CHECKED 2026-05-27

NEED 2 to 6 scopes · SPARE 0 · BUY 2 · ON HAND unknown · LOCAL BETTER yes

PICK USB digital microscope 1000x search · FIND Amazon search: usb digital microscope 1000x · CHEAPER Borrow  
microscopes

PURPOSE View and count stomata

STORAGE Microscope case

SOURCE <https://observer.globe.gov/do-globe-observer/trees>

NOTES Borrow strongly preferred.



**JUNAN 2mm x 1mm rare earth disc magnets, 100 pack**

REQUIRED

\$6.99 UNIT

\$13.98 TOTAL

PY-STEM · PYS-01 · MAGNETIC CAPSULE MAZE CUP · MAGNETS · REUSABLE · CHECKED 2026-05-27

NEED 12 to 18 magnets · SPARE 15 percent extra · BUY 2 · ON HAND no · LOCAL BETTER no

PICK JUNAN 2mm x 1mm Tiny Rare Earth Disc Magnets 100

Pack · FIND [amazon.com/dp/B09V14FGQF](https://www.amazon.com/dp/B09V14FGQF) · ASIN B09V14FGQF · CHEAPER Teacher magnet stash

PURPOSE External magnetic control

SAFETY Choking hazard. Keep away from mouths, electronics, and medical implants.

STORAGE Magnet case

SOURCE <https://www.sciencemuseumgroup.org.uk/learning/resources/magnetic-maze>

NOTES Verified public Amazon price in current search.

**6mm x 1.5mm neodymium disc magnets, 100 pack**

RECOMMENDED

\$19.49 UNIT

\$19.49 TOTAL

PY-STEM · PYS-01 · MAGNETIC CAPSULE MAZE CUP · MAGNETS · REUSABLE · CHECKED 2026-05-27

NEED 6 to 12 magnets · SPARE 15 percent extra · BUY 1 · ON HAND no · LOCAL BETTER no

PICK Small Neodymium Disc Magnets 6x1.5mm 100

Pack · FIND [amazon.com/dp/B0BV6P6YYS](https://www.amazon.com/dp/B0BV6P6YYS) · ASIN B0BV6P6YYS · CHEAPER Ferrite magnets if strong enough

PURPOSE Stronger external control option

SAFETY Choking hazard. Count in and out.

STORAGE Magnet case

SOURCE <https://www.sciencemuseumgroup.org.uk/learning/resources/magnetic-maze>

NOTES Verified public Amazon price in current search.

**Maze boards, barrier sheets, steel tokens**

REQUIRED

\$10 UNIT

\$10 TOTAL

LOCAL ESTIMATE

PY-STEM · PYS-01 · MAGNETIC CAPSULE MAZE CUP · PRINTING/BUILD · REUSABLE · CHECKED 2026-05-27

NEED 6 boards · SPARE 1 spare · BUY 1 · ON HAND no · LOCAL BETTER yes

PICK local print and plastic sleeves · FIND local purchase · CHEAPER Laminate paper mazes

PURPOSE Maze simulation

SAFETY No magnets in pockets.

STORAGE Magnetic maze bin

SOURCE <https://www.sciencemuseumgroup.org.uk/learning/resources/magnetic-maze>

NOTES Table-based.

**Cornstarch and zip bags**

REQUIRED

\$18 UNIT

\$18 TOTAL

LOCAL ESTIMATE

PY-STEM · PYS-02 · OOBLECK ARMOR ARENA · GROCERY · CONSUMABLE · CHECKED 2026-05-27

NEED 6 team stations · SPARE 15 percent extra · BUY 1 · ON HAND no · LOCAL BETTER yes

PICK cornstarch bulk and zip bags · FIND Amazon search: cornstarch bulk zip bags · CHEAPER Local grocery

PURPOSE Non-Newtonian armor pads

SAFETY Corn allergy note. No eating. Double-bag.

STORAGE Oobleck bin

SOURCE <https://www.sciencebuddies.org/stem-activities/oobleck-colloid>

NOTES Clean immediately.

## Cardboard, skewers, straws, craft sticks

REQUIRED

\$15 UNIT

\$15 TOTAL

LOCAL ESTIMATE

PY-STEM · PYS-03 · CARDBOARD AUTOMATA ARCADE · BUILD SUPPLIES · CONSUMABLE · CHECKED 2026-05-27

**NEED** 6 team kits · **SPARE** 15 percent extra · **BUY** 1 · **ON HAND** maybe yes · **LOCAL BETTER** yes

**PICK** craft sticks skewers straws bulk · **FIND** Amazon search: craft sticks skewers straws bulk · **CHEAPER** Reuse cardboard

**PURPOSE** Cam and follower automata

**SAFETY** Hot glue staff-supervised.

**STORAGE** Automata bin

**SOURCE** <https://www.exploratorium.edu/tinkering/projects/cardboard-automata>

**NOTES** Prep sample cam.

## Plastic funnels and flexible tubing

RECOMMENDED

\$12 UNIT

\$48 TOTAL

UNVERIFIED

PY-STEM · PYS-04 · STETHOSCOPE SPRINT AND RECOVERY CHALLENGE · TUBING/FUNNELS · REUSABLE · CHECKED 2026-05-27

**NEED** 6 to 12 funnels, 30 ft tubing · **SPARE** 15 percent extra · **BUY** 4 · **ON HAND** maybe no · **LOCAL BETTER** yes

**PICK** Norpro Plastic Funnel Set of 3 and vinyl

tubing · **FIND** [amazon.com/dp/B000HJBFC6](https://www.amazon.com/dp/B000HJBFC6) · **ASIN** B000HJBFC6 · **CHEAPER** Dollar-store funnels and hardware-store tubing

**PURPOSE** Homemade stethoscope

**SAFETY** Sanitize shared parts. No forced heart-rate measurement.

**STORAGE** Biomedical bin

**SOURCE** <https://www.sciencebuddies.org/stem-activities/make-stethoscope>

**NOTES** Price hidden or not confirmed.

## Sanitizing wipes

REQUIRED

\$8 UNIT

\$8 TOTAL

LOCAL ESTIMATE

PY-STEM · PYS-04 · STETHOSCOPE SPRINT AND RECOVERY CHALLENGE · SANITATION · CONSUMABLE · CHECKED 2026-05-27

**NEED** 1 tub · **SPARE** 1 spare if available · **BUY** 1 · **ON HAND** maybe yes · **LOCAL BETTER** yes

**PICK** disinfecting wipes classroom · **FIND** Amazon search: disinfecting wipes classroom · **CHEAPER** School nurse supply

**PURPOSE** Shared device cleaning

**SAFETY** Follow site policy.

**STORAGE** Safety bin

**SOURCE** <https://www.sciencebuddies.org/stem-activities/make-stethoscope>

**NOTES** Safety-critical.

## Meter sticks, cones, score sheets

RECOMMENDED

\$10 UNIT

\$10 TOTAL

LOCAL ESTIMATE

PY-STEM · PYS-05 · REACTION TIME COMBINE · GYM SUPPLIES · REUSABLE · CHECKED 2026-05-27

**NEED** 6 stations · **SPARE** 1 spare meter stick · **BUY** 1 · **ON HAND** maybe yes · **LOCAL BETTER** yes

**PICK** meter sticks and gym cones · **FIND** Amazon search: meter sticks gym cones · **CHEAPER** Borrow from PE/science

**PURPOSE** Reaction time stations

**SAFETY** No sprint collisions. Clear lanes.

**STORAGE** Gym bin

**SOURCE** <https://www.sciencebuddies.org/stem-activities/think-fast-measure-your-reaction-time>

**NOTES** Can be fully seated.

## Original Slinky Jr or metal slinkies

RECOMMENDED

\$6 UNIT

\$36 TOTAL

UNVERIFIED

PY-STEM · PYS-06 · SONAR SLINKY SHOWDOWN · WAVES TOOL · REUSABLE · CHECKED 2026-05-27

**NEED** 4 to 6 slinkies · **SPARE** 1 spare · **BUY** 6 · **ON HAND** maybe no · **LOCAL BETTER** yes

**PICK** Original Slinky Jr · **FIND** [amazon.com/dp/B000000IZGI](https://www.amazon.com/dp/B000000IZGI) · **ASIN** B000000IZGI · **CHEAPER** Borrow from toy/science closet

**PURPOSE** Longitudinal wave modeling

**SAFETY** Finger pinch warning. Do not overstretch.

**STORAGE** Wave bin

**SOURCE** <https://classroom.iihs.org/sonar-slinky/>

**NOTES** Price hidden or sign-in gated.

## Cardstock, foil, tape, paper screens

REQUIRED

\$8 UNIT

\$8 TOTAL

LOCAL ESTIMATE

PY-STEM · PYS-07 · PINHOLE PRECISION CHALLENGE · OPTICS BUILD · CONSUMABLE · CHECKED 2026-05-27

**NEED** 6 team kits · **SPARE** 15 percent extra · **BUY** 1 · **ON HAND** maybe yes · **LOCAL BETTER** yes

**PICK** aluminum foil cardstock tape · **FIND** Amazon search: aluminum foil cardstock tape · **CHEAPER** Local grocery and school supply

**PURPOSE** Pinhole image viewer

**SAFETY** No direct sun viewing. Pins controlled by staff.

**STORAGE** Optics bin

**SOURCE** <https://www.exploratorium.edu/snacks/personal-pinhole-theater>

**NOTES** Indoor light source works.

## Force-map cards and clipboards

REQUIRED

\$10 UNIT

\$10 TOTAL

LOCAL ESTIMATE

PY-STEM · PYS-08 · LOW-ROPES FORCE MAP RELAY · PRINTING · REUSABLE · CHECKED 2026-05-27

**NEED** 6 team sets · **SPARE** 1 spare · **BUY** 1 · **ON HAND** no · **LOCAL BETTER** yes

**PICK** local print and laminate · **FIND** local purchase · **CHEAPER** Black and white cards

**PURPOSE** Ropes-course physics wraparound

**SAFETY** Host ropes rules override module.

**STORAGE** Ambler trip folder

**SOURCE** [https://www.teachengineering.org/activities/view/wpi\\_empathy\\_activity3](https://www.teachengineering.org/activities/view/wpi_empathy_activity3)

**NOTES** Coordinate with Holly.

## Balloons, pop-top caps, recycled CDs

REQUIRED

\$20 UNIT

\$20 TOTAL

LOCAL ESTIMATE

PY-STEM · PYS-09 · HOVERCRAFT HOCKEY HACKATHON · BUILD SUPPLIES · MIXED · CHECKED 2026-05-27

**NEED** 12 to 18 hovercraft kits · **SPARE** 15 percent extra · **BUY** 1 · **ON HAND** maybe no · **LOCAL BETTER** yes

**PICK** latex balloons pop top bottle caps · **FIND** Amazon search: latex balloons pop top bottle caps · **CHEAPER** Local party store, collect caps/CDs

**PURPOSE** Mini hovercraft build

**SAFETY** Latex allergy note. No face-level launches.

**STORAGE** Hovercraft bin

**SOURCE** <https://www.jpl.nasa.gov/edu/resources/lesson-plan/hovering-on-a-cushion-of-air/>

**NOTES** Use nonlatex fallback if needed.

## Diffraction glasses or gratings

REQUIRED

\$30 UNIT

\$30 TOTAL

UNVERIFIED

PY-STEM · PYS-10 · SPECTRA SLEUTH SHOWDOWN · OPTICS · REUSABLE · CHECKED 2026-05-27

NEED 30 to 50 pairs · SPARE 5 spare · BUY 1 · ON HAND no · LOCAL BETTER maybe

PICK Fireworks Diffraction Glasses 51 total · FIND amazon.com/dp/B00JV06CV6 · ASIN B00JV06CV6 · CHEAPER Cut diffraction film strips

PURPOSE Safe spectra viewing

SAFETY Novelty eyewear only. Not eye protection. No lasers.

STORAGE Spectra envelope

SOURCE <https://www.sciencehistory.org/visit/exhibitions/history-of-fireworks/>

NOTES No flame tests.

## LED and safe classroom light sources

RECOMMENDED

\$20 UNIT

\$20 TOTAL

LOCAL ESTIMATE

PY-STEM · PYS-10 · SPECTRA SLEUTH SHOWDOWN · LIGHT SOURCES · REUSABLE · CHECKED 2026-05-27

NEED 6 stations · SPARE 1 spare · BUY 1 · ON HAND maybe yes · LOCAL BETTER yes

PICK LED flashlight color light set · FIND Amazon search: led flashlight color light set · CHEAPER Borrow lamps/LEDs

PURPOSE Mystery spectra stations

SAFETY No lasers. Avoid hot bulbs.

STORAGE Spectra bin

SOURCE <https://www.sciencehistory.org/visit/exhibitions/history-of-fireworks/>

NOTES Use safe lights.

## Bin cards, address labels, order deck, route mat

REQUIRED

\$20 UNIT

\$20 TOTAL

LOCAL ESTIMATE

PY-STEM · PYS-11 · BOOKBOT BIN LOGIC CHALLENGE · PRINTING/BUILD · REUSABLE · CHECKED 2026-05-27

NEED 6 sets · SPARE 1 spare · BUY 1 · ON HAND no · LOCAL BETTER yes

PICK local print and laminate · FIND local purchase · CHEAPER Index cards and painter tape

PURPOSE ASRS addressing simulation

SAFETY Keep clear of actual library operations.

STORAGE BookBot folder

SOURCE <https://library.temple.edu/spaces/charles-in-charge>

NOTES Frame as bin addressing and search, not throughput.

## Foam board, dowels, binder clips, toy cart

REQUIRED

\$25 UNIT

\$25 TOTAL

LOCAL ESTIMATE

PY-STEM · PYS-12 · ACCESSIBILITY RAMP RESCUE LAB · BUILD SUPPLIES · MIXED · CHECKED 2026-05-27

NEED 6 team kits · SPARE 15 percent extra · BUY 1 · ON HAND maybe no · LOCAL BETTER yes

PICK foam board dowels binder clips toy cart · FIND Amazon search: foam board dowels binder clips toy cart · CHEAPER Borrow toy cars, use cardboard

PURPOSE Portable ramp model

SAFETY Keep loads low. No standing on models.

STORAGE Accessibility bin

SOURCE [https://www.teachengineering.org/activities/view/wpi\\_empathy\\_activity3](https://www.teachengineering.org/activities/view/wpi_empathy_activity3)

NOTES ADA-style slope model.

**Vinyl STEM stickers**

PRIZE

\$10 UNIT

\$20 TOTAL

LOCAL ESTIMATE

SHARED · REWARDS · PRIZE TABLE · PRIZE · CONSUMABLE · CHECKED 2026-05-27

NEED 75 to 100 · SPARE 15 percent extra · BUY 2 · ON HAND no · LOCAL BETTER yes

PICK STEM stickers 100 pack · FIND Amazon search: stem stickers 100 pack · CHEAPER Local school store

PURPOSE 1 to 2 ticket prize

STORAGE Prize bin

SOURCE planning line

NOTES Non-candy.

**Mechanical pencils and gel pens**

PRIZE

\$12 UNIT

\$24 TOTAL

LOCAL ESTIMATE

SHARED · REWARDS · PRIZE TABLE · PRIZE · CONSUMABLE · CHECKED 2026-05-27

NEED 50 to 75 · SPARE 15 percent extra · BUY 2 · ON HAND no · LOCAL BETTER yes

PICK mechanical pencils gel pens bulk · FIND Amazon search: mechanical pencils gel pens bulk · CHEAPER School supply closet

PURPOSE 1 to 4 ticket prize

STORAGE Prize bin

SOURCE planning line

NOTES Useful prize.

**Mini notebooks**

PRIZE

\$15 UNIT

\$15 TOTAL

LOCAL ESTIMATE

SHARED · REWARDS · PRIZE TABLE · PRIZE · CONSUMABLE · CHECKED 2026-05-27

NEED 30 · SPARE 15 percent extra · BUY 1 · ON HAND no · LOCAL BETTER yes

PICK mini notebooks bulk · FIND Amazon search: mini notebooks bulk · CHEAPER Local dollar store

PURPOSE 2 to 4 ticket prize

STORAGE Prize bin

SOURCE planning line

NOTES Non-candy.

**Fidget puzzles or mini desk puzzles**

PRIZE

\$18 UNIT

\$18 TOTAL

LOCAL ESTIMATE

SHARED · REWARDS · PRIZE TABLE · PRIZE · CONSUMABLE · CHECKED 2026-05-27

NEED 20 to 30 · SPARE 15 percent extra · BUY 1 · ON HAND no · LOCAL BETTER yes

PICK mini desk puzzles bulk · FIND Amazon search: mini desk puzzles bulk · CHEAPER Local store

PURPOSE 4 to 7 ticket prize

SAFETY Avoid sharp or weapon-like toys.

STORAGE Prize bin

SOURCE planning line

NOTES Middle-school appropriate.

Diffraction glasses, extra pairs

PRIZE

\$30 UNIT

\$30 TOTAL

UNVERIFIED

SHARED · REWARDS · PRIZE TABLE · PRIZE · REUSABLE · CHECKED 2026-05-27

NEED 10 to 20 · SPARE 5 spare · BUY 1 · ON HAND no · LOCAL BETTER maybe

PICK Fireworks Diffraction Glasses 51 total · FIND amazon.com/dp/B00JV06CV6 · ASIN B00JV06CV6 · CHEAPER Sticker bundles

PURPOSE Spectra tie-in and prize

SAFETY Novelty eyewear only. Not PPE.

STORAGE Prize bin

SOURCE <https://www.sciencehistory.org/visit/exhibitions/history-of-fireworks/>

NOTES Shared with PYS-10 if enough pairs.

**Safety goggles, class set**

REQUIRED

\$30 UNIT

\$30 TOTAL

LOCAL ESTIMATE

SHARED · ALL · ALL ACTIVITIES · PPE · REUSABLE · CHECKED 2026-05-27

NEED 30 goggles · SPARE 0 if existing · BUY 1 · ON HAND maybe yes · LOCAL BETTER yes

PICK classroom safety goggles 30 pack · FIND Amazon search: classroom safety goggles 30 pack · CHEAPER Borrow from science stockroom

PURPOSE Eye protection for splash, soil, plant peel, and oobleck activities

SAFETY Sanitize between users.

STORAGE PPE bin

SOURCE ACS/NSTA general lab safety

NOTES Safety-critical.

**Nitrile gloves, 100 pack**

REQUIRED

\$12 UNIT

\$24 TOTAL

LOCAL ESTIMATE

SHARED · ALL · ALL ACTIVITIES · PPE · CONSUMABLE · CHECKED 2026-05-27

NEED 2 boxes · SPARE 15 percent built into second box · BUY 2 · ON HAND maybe yes · LOCAL BETTER yes

PICK nitrile gloves medium 100 pack · FIND Amazon search: nitrile gloves medium 100 pack · CHEAPER School stockroom

PURPOSE Soil, plant, oobleck, and cleanup handling

SAFETY Latex-free preferred.

STORAGE PPE bin

SOURCE ACS/NSTA general lab safety

NOTES Safety-critical.

**First aid kit and incident log binder**

REQUIRED

\$25 UNIT

\$25 TOTAL

LOCAL ESTIMATE

SHARED · ALL · ALL ACTIVITIES · SAFETY · REUSABLE · CHECKED 2026-05-27

NEED 1 · SPARE 0 · BUY 1 · ON HAND maybe yes · LOCAL BETTER yes

PICK first aid kit classroom · FIND Amazon search: classroom first aid kit · CHEAPER School nurse / camp office

PURPOSE Emergency response and documentation

SAFETY Do not replace official site procedures.

STORAGE Safety cabinet

SOURCE planning line

NOTES Safety-critical.

**Sunscreen, water station, weather plan signs**

REQUIRED

\$25 UNIT

\$25 TOTAL

LOCAL ESTIMATE

SHARED · ALL · OUTDOOR MODULES · SAFETY · CONSUMABLE · CHECKED 2026-05-27

NEED 1 set · SPARE extra water · BUY 1 · ON HAND maybe yes · LOCAL BETTER yes

PICK local purchase · FIND local purchase · CHEAPER Camp office supply

PURPOSE Outdoor safety support

SAFETY Allergy-sensitive sunscreen policy.

STORAGE Outdoor safety bin

SOURCE planning line

NOTES Confirm site policy.

Student handout packet printing

REQUIRED

\$75 UNIT

\$75 TOTAL

LOCAL ESTIMATE

SHARED · PRINTING · ALL PACKETS · PRINTING · CONSUMABLE · CHECKED 2026-05-27

NEED 50 packets or as needed · SPARE 15 percent extra · BUY 1 · ON HAND no · LOCAL BETTER yes

PICK local print · FIND local purchase · CHEAPER Black and white with spot color

PURPOSE Camp handout packets and individual PDFs

STORAGE Print folder

SOURCE planning line

NOTES Estimate only.

Station signs, score slips, tokens, certificates

REQUIRED

\$60 UNIT

\$60 TOTAL

LOCAL ESTIMATE

SHARED · PRINTING · ALL SIGNS · PRINTING · CONSUMABLE · CHECKED 2026-05-27

NEED 500 pages · SPARE 15 percent extra · BUY 1 · ON HAND no · LOCAL BETTER yes

PICK local print · FIND local purchase · CHEAPER Use cardstock only for durable items

PURPOSE Printable competition kit

STORAGE Print folder

SOURCE planning line

NOTES Estimate only.



## trays, soil, string or yarn roots, craft sticks

BACKUP-ONLY

\$12 UNIT

\$12 TOTAL

LOCAL ESTIMATE

FROM TREES TO TECH · TTB-01 · ROOT GRIP EROSION RESCUE · BACKUP KIT · MIXED · CHECKED 2026-05-27

**NEED** 6 team kits · **SPARE** 15 percent extra · **BUY** 1 · **ON HAND** maybe yes · **LOCAL BETTER** yes

**PICK** Amazon search for listed materials · **FIND** Amazon search: trays soil string or yarn roots · **CHEAPER** Borrow or use existing supplies

**PURPOSE** Backup-only activity materials

**SAFETY** Use trays and wipe spills.

**STORAGE** TTB-01 backup bin

**SOURCE** EPA green infrastructure and erosion lessons

**NOTES** Backup-only. Do not order unless needed.

## clinometers, string, washers, rulers

BACKUP-ONLY

\$12 UNIT

\$12 TOTAL

LOCAL ESTIMATE

FROM TREES TO TECH · TTB-02 · TREE HEIGHT TRIANGULATION SHOOTOUT · BACKUP KIT · MIXED · CHECKED 2026-05-27

**NEED** 6 team kits · **SPARE** 15 percent extra · **BUY** 1 · **ON HAND** maybe yes · **LOCAL BETTER** yes

**PICK** Amazon search for listed materials · **FIND** Amazon search: clinometers string washers · **CHEAPER** Borrow or use existing supplies

**PURPOSE** Backup-only activity materials

**SAFETY** Outdoor boundaries.

**STORAGE** TTB-02 backup bin

**SOURCE** GLOBE Trees Activities

**NOTES** Backup-only. Do not order unless needed.

## thermometers or IR thermometers, maps, clipboards

BACKUP-ONLY

\$12 UNIT

\$12 TOTAL

LOCAL ESTIMATE

FROM TREES TO TECH · TTB-03 · URBAN HEAT SHADE DASH · BACKUP KIT · MIXED · CHECKED 2026-05-27

**NEED** 6 team kits · **SPARE** 15 percent extra · **BUY** 1 · **ON HAND** maybe yes · **LOCAL BETTER** yes

**PICK** Amazon search for listed materials · **FIND** Amazon search: thermometers or IR thermometers maps clipboards · **CHEAPER** Borrow or use existing supplies

**PURPOSE** Backup-only activity materials

**SAFETY** Check heat index. Walk, do not run.

**STORAGE** TTB-03 backup bin

**SOURCE** NASA MyNASADData Urban Heat Island

**NOTES** Backup-only. Do not order unless needed.

## spinach leaves, baking soda solution, cups, oral syringes without needles

BACKUP-ONLY

\$12 UNIT

\$12 TOTAL

LOCAL ESTIMATE

FROM TREES TO TECH · TTB-04 · PHOTOSYNTHESIS FLOAT-OFF PLAYOFFS · BACKUP KIT · MIXED · CHECKED 2026-05-27

**NEED** 6 team kits · **SPARE** 15 percent extra · **BUY** 1 · **ON HAND** maybe yes · **LOCAL BETTER** yes

**PICK** Amazon search for listed materials · **FIND** Amazon search: spinach leaves baking soda solution cups · **CHEAPER** Borrow or use existing supplies

**PURPOSE** Backup-only activity materials

**SAFETY** No tasting. Handle syringes as lab tools.

**STORAGE** TTB-04 backup bin

**SOURCE** Exploratorium Photosynthetic Floatation

**NOTES** Backup-only. Do not order unless needed.

## pencils, foam, cardboard, weights

BACKUP-ONLY

\$12 UNIT

\$12 TOTAL

LOCAL ESTIMATE

PY-STEM · PYB-01 · PENCIL PRESSURE SAFETY LAB · BACKUP KIT · MIXED · CHECKED 2026-05-27

**NEED** 6 team kits · **SPARE** 15 percent extra · **BUY** 1 · **ON HAND** maybe yes · **LOCAL BETTER** yes

**PICK** Amazon search for listed materials · **FIND** Amazon search: pencils foam cardboard · **CHEAPER** Borrow or use existing supplies

**PURPOSE** Backup-only activity materials

**SAFETY** No skin tests. Use foam only.

**STORAGE** PYB-01 backup bin

**SOURCE** IIHS Pencil Pressure

**NOTES** Backup-only. Do not order unless needed.

## dominoes, rulers, tape, index cards

BACKUP-ONLY

\$12 UNIT

\$12 TOTAL

LOCAL ESTIMATE

PY-STEM · PYB-02 · DOMINO NEURON RELAY · BACKUP KIT · MIXED · CHECKED 2026-05-27

**NEED** 6 team kits · **SPARE** 15 percent extra · **BUY** 1 · **ON HAND** maybe yes · **LOCAL BETTER** yes

**PICK** Amazon search for listed materials · **FIND** Amazon search: dominoes rulers tape · **CHEAPER** Borrow or use existing supplies

**PURPOSE** Backup-only activity materials

**SAFETY** Low risk.

**STORAGE** PYB-02 backup bin

**SOURCE** Exploratorium Domino Effect

**NOTES** Backup-only. Do not order unless needed.

## classroom pulleys, string, weights, spring scales if available

BACKUP-ONLY

\$12 UNIT

\$12 TOTAL

LOCAL ESTIMATE

PY-STEM · PYB-03 · PULLEY RESCUE RELAY · BACKUP KIT · MIXED · CHECKED 2026-05-27

**NEED** 6 team kits · **SPARE** 15 percent extra · **BUY** 1 · **ON HAND** maybe yes · **LOCAL BETTER** yes

**PICK** Amazon search for listed materials · **FIND** Amazon search: classroom pulleys string weights · **CHEAPER** Borrow or use existing supplies

**PURPOSE** Backup-only activity materials

**SAFETY** Keep feet clear of loads.

**STORAGE** PYB-03 backup bin

**SOURCE** TryEngineering Pulleys and Force

**NOTES** Backup-only. Do not order unless needed.

printed barcode cards, dry erase boards,  
timers

BACKUP-ONLY

\$12 UNIT

\$12 TOTAL

LOCAL ESTIMATE

PY-STEM · PYB-04 · BARCODE CHECKSUM RESCUE · BACKUP KIT · MIXED · CHECKED 2026-05-27

NEED 6 team kits · SPARE 15 percent extra · BUY 1 · ON HAND maybe yes · LOCAL BETTER yes

PICK Amazon search for listed materials · FIND Amazon search: printed barcode cards dry erase boards  
timers · CHEAPER Borrow or use existing supplies

PURPOSE Backup-only activity materials

SAFETY Normal classroom management.

STORAGE PYB-04 backup bin

SOURCE Computer science unplugged error-detection concepts

NOTES Backup-only. Do not order unless needed.

## Fresh soil or mud samples

REQUIRED

LOCAL ESTIMATE

FROM TREES TO TECH · TTT-01 · MUDWATT BIOELECTRIC LEAGUE · CONSUMABLE · CONSUMABLE · CHECKED 2026-05-27

NEED 6 sealed containers · SPARE 15 percent extra · BUY 1 · ON HAND no · LOCAL BETTER yes

PICK local soil or potting soil · FIND local purchase or collection · CHEAPER Collect from approved campus site

PURPOSE Microbial fuel source

SAFETY Avoid contaminated soil. Gloves required.

STORAGE Sealed soil tub

SOURCE <https://www.magicalmicrobes.com/pages/how-mudwatt-works>

NOTES Use approved campus collection only.

## Mini hygrometer/thermometer units

RECOMMENDED

FROM TREES TO TECH · TTT-02 · FOREST SENSOR SPRINT · FIELD MEASUREMENT · REUSABLE · CHECKED 2026-05-27

NEED 6 to 8 units · SPARE 1 spare · BUY 2 · ON HAND no · LOCAL BETTER yes

PICK mini digital hygrometer thermometer · FIND Amazon search: mini digital hygrometer thermometer · CHEAPER Borrow from lab

PURPOSE Microclimate data collection

STORAGE Sensor bin

SOURCE <https://observer.globe.gov/do-globe-observer/trees>

NOTES Use relative comparisons if readings drift.

## 3-in-1 soil moisture meter

RECOMMENDED

UNVERIFIED

FROM TREES TO TECH · TTT-02 · FOREST SENSOR SPRINT · FIELD MEASUREMENT · REUSABLE · CHECKED 2026-05-27

NEED 6 units · SPARE 0 to 1 spare · BUY 6 · ON HAND no · LOCAL BETTER yes

PICK SONKIR Soil pH Meter MS02 3-in-1 · FIND [amazon.com/dp/B07BR52P26](https://www.amazon.com/dp/B07BR52P26) · ASIN B07BR52P26 · CHEAPER Moisture-only meter

PURPOSE Optional soil comparison

SAFETY Probe tips are pointed.

STORAGE Wipe probes clean

SOURCE <https://observer.globe.gov/do-globe-observer/trees>

NOTES Price hidden or not confirmed.

## Paper clinometer templates

REQUIRED

LOCAL ESTIMATE

FROM TREES TO TECH · TTT-02 · FOREST SENSOR SPRINT · PRINTING · CONSUMABLE · CHECKED 2026-05-27

NEED 30 sheets · SPARE 5 spare · BUY 1 · ON HAND no · LOCAL BETTER yes

PICK local print · FIND local purchase · CHEAPER Hand-draw masters

PURPOSE Tree height estimates

STORAGE Field folder

SOURCE <https://observer.globe.gov/do-globe-observer/trees>

NOTES Use GLOBE-style clinometer.

## Paper, cardstock, paper clips, string

REQUIRED

FROM TREES TO TECH · TTT-03 · SEED DISPERSAL DERBY · BUILD SUPPLIES · CONSUMABLE · CHECKED 2026-05-27

NEED 6 team kits · SPARE 15 percent extra · BUY 1 · ON HAND maybe yes · LOCAL BETTER yes

PICK cardstock paper clips string bundle · FIND Amazon search: cardstock paper clips string · CHEAPER Use existing office supplies

PURPOSE Passive seed prototypes

SAFETY No throwing at people. Low-power fan only.

STORAGE Seed bin

SOURCE <https://www.sciencebuddies.org/stem-activities/gone-with-the-wind-plant-seed-dispersal>

NOTES No paper airplane or rocket builds.

## Paper towels, cotton string, cups, trays

REQUIRED

FROM TREES TO TECH · TTT-04 · XYLEM PIPELINE RELAY · LAB SUPPLIES · CONSUMABLE · CHECKED 2026-05-27

NEED 6 tray stations · SPARE 15 percent extra · BUY 1 · ON HAND maybe yes · LOCAL BETTER yes

PICK paper towels cotton string plastic cups trays · FIND Amazon search: paper towels cotton string plastic cups trays · CHEAPER Dollar store or school stock

PURPOSE Capillary transport race

SAFETY Use trays. Wipe spills immediately.

STORAGE Water activity bin

SOURCE <https://www.sciencebuddies.org/stem-activities/gone-with-the-wind-plant-seed-dispersal>

NOTES Use washable dye only.

## Plant profile cards and control dial boards

REQUIRED

\$15 UNIT

\$15 TOTAL

LOCAL ESTIMATE

FROM TREES TO TECH · TTT-05 · GREENHOUSE CLIMATE CONTROLLER CHALLENGE · PRINTING · REUSABLE · CHECKED 2026-05-27

NEED 6 laminated sets · SPARE 1 spare set · BUY 1 · ON HAND no · LOCAL BETTER yes

PICK local print and laminate · FIND local purchase · CHEAPER Sheet protectors instead of lamination

PURPOSE Greenhouse settings game

SAFETY Stay on tour route. Do not touch restricted controls.

STORAGE Greenhouse folder

SOURCE <https://ambler.temple.edu/campus-resources/greenhouse>

NOTES Use actual tour evidence.

## Dry pine cones

RECOMMENDED

LOCAL ESTIMATE

FROM TREES TO TECH · TTT-06 · PINECONE WEATHER MACHINE CHALLENGE · NATURAL MATERIAL · REUSABLE · CHECKED 2026-05-27

NEED 12 to 18 cones · SPARE 3 spare · BUY 1 · ON HAND maybe no · LOCAL BETTER yes

PICK local nature collection · FIND local collection · CHEAPER Ask grounds crew

PURPOSE Humidity-responsive device

SAFETY Inspect for insects, sap, and sharp edges.

STORAGE Dry natural-materials bin

SOURCE <https://asknature.org/strategy/pine-cones-open-and-close-in-response-to-weather/>

NOTES Dry before use.

## Spray bottles, index cards, skewers, rulers

REQUIRED

FROM TREES TO TECH · TTT-06 · PINECONE WEATHER MACHINE CHALLENGE · SUPPLIES · MIXED · CHECKED 2026-05-27

NEED 6 stations · SPARE 15 percent extra · BUY 1 · ON HAND maybe yes · LOCAL BETTER yes

PICK spray bottles index cards skewers rulers · FIND Amazon search: spray bottles index cards skewers rulers · CHEAPER Use existing classroom supplies

PURPOSE Weather machine build

SAFETY Keep floors dry after spray.

STORAGE Weather machine bin

SOURCE <https://asknature.org/strategy/pine-cones-open-and-close-in-response-to-weather/>

NOTES Table-based.

## Plant and pollinator card decks

REQUIRED

\$25 UNIT

\$25 TOTAL

LOCAL ESTIMATE

FROM TREES TO TECH · TTT-07 · POLLINATOR NETWORK DRAFT AND BUILD · PRINTING · REUSABLE · CHECKED 2026-05-27

NEED 6 decks · SPARE 1 spare deck · BUY 1 · ON HAND no · LOCAL BETTER yes

PICK local print and laminate · FIND local purchase · CHEAPER Sheet protectors

PURPOSE Pollinator habitat strategy game

SAFETY No live insects.

STORAGE Pollinator folder

SOURCE <https://www.usda.gov/farming-and-ranching/plants-and-crops/plant-breeding/pollinators>

NOTES Print in color if possible.

## Clue cards, maps, evidence tokens

REQUIRED

\$20 UNIT

\$20 TOTAL

LOCAL ESTIMATE

FROM TREES TO TECH · TTT-08 · ARBORETUM ECO-QUEST · PRINTING · REUSABLE · CHECKED 2026-05-27

NEED 6 route packs · SPARE 1 spare pack · BUY 1 · ON HAND no · LOCAL BETTER yes

PICK local print and laminate · FIND local purchase · CHEAPER Black and white route cards

PURPOSE Outdoor checkpoint challenge

SAFETY Buddy system. Approved route only.

STORAGE Arboretum folder

SOURCE <https://observer.globe.gov/do-globe-observer/trees>

NOTES Confirm route with host.

## Minecraft Education devices and world template

REQUIRED

FROM TREES TO TECH · TTT-09 · MINECRAFT TREE WORLD RESILIENCE CUP · SOFTWARE · REUSABLE · CHECKED 2026-05-27

NEED 6 devices plus projector · SPARE 0 · BUY 0 · ON HAND unknown · LOCAL BETTER yes

PICK Minecraft Education licenses · FIND [education.minecraft.net/en-us](https://education.minecraft.net/en-us) · CHEAPER Paper resilience design board

PURPOSE Digital ecosystem build

SAFETY Standard device use and login management.

STORAGE Digital kit

SOURCE <https://education.minecraft.net/en-us/resources/science/biodiversity>

NOTES Confirm licensing.

## Tree ring image cards and answer boards

REQUIRED

\$20 UNIT

\$20 TOTAL

LOCAL ESTIMATE

FROM TREES TO TECH · TTT-10 · TREE RING CLIMATE DETECTIVE TOURNAMENT · PRINTING · REUSABLE · CHECKED 2026-05-27

NEED 6 sets · SPARE 1 spare · BUY 1 · ON HAND no · LOCAL BETTER yes

PICK local print and laminate · FIND local purchase · CHEAPER Wood cookies if already owned

PURPOSE Climate proxy data tournament

SAFETY Inspect wood slices for splinters if used.

STORAGE Tree ring folder

SOURCE <https://www.ncei.noaa.gov/products/paleoclimatology/tree-ring>

NOTES Use printed images for safer reset.

## Pipettes, wax paper, sandpaper, ramps

REQUIRED

FROM TREES TO TECH · TTT-11 · LOTUS LEAF SURFACE SPRINT · LAB SUPPLIES · MIXED · CHECKED 2026-05-27

NEED 6 team sets · SPARE 15 percent extra · BUY 1 · ON HAND maybe no · LOCAL BETTER yes

PICK transfer pipettes wax paper sandpaper · FIND Amazon search: transfer pipettes wax paper sandpaper · CHEAPER Science stockroom

PURPOSE Hydrophobic surface tests

SAFETY Keep floors dry. Washable dirt proxy only.

STORAGE Surface sprint bin

SOURCE <https://asknature.org/strategy/surface-allows-self-cleaning/>

NOTES No major staining.

## Clear nail polish, tape, leaf samples

REQUIRED

FROM TREES TO TECH · TTT-12 · LEAF STOMATA MICROSCOPE DETECTIVE · LAB SUPPLIES · CONSUMABLE · CHECKED 2026-05-27

NEED 6 team sets · SPARE 15 percent extra · BUY 1 · ON HAND no · LOCAL BETTER yes

PICK clear nail polish tape microscope slides · FIND Amazon search: clear nail polish tape microscope slides · CHEAPER Use phone macro lens and prepared peels

PURPOSE Leaf peel microscopy detective activity

SAFETY Ventilation for nail polish. Gloves optional.

STORAGE Stomata bin

SOURCE <https://observer.globe.gov/do-globe-observer/trees>

NOTES Legacy-safe replacement.

## USB digital microscope or classroom microscopes

RECOMMENDED

FROM TREES TO TECH · TTT-12 · LEAF STOMATA MICROSCOPE DETECTIVE · EQUIPMENT · REUSABLE · CHECKED 2026-05-27

NEED 2 to 6 scopes · SPARE 0 · BUY 2 · ON HAND unknown · LOCAL BETTER yes

PICK USB digital microscope 1000x search · FIND Amazon search: usb digital microscope 1000x · CHEAPER Borrow microscopes

PURPOSE View and count stomata

STORAGE Microscope case

SOURCE <https://observer.globe.gov/do-globe-observer/trees>

NOTES Borrow strongly preferred.

## Maze boards, barrier sheets, steel tokens

REQUIRED

\$10 UNIT

\$10 TOTAL

LOCAL ESTIMATE

PY-STEM • PYS-01 • MAGNETIC CAPSULE MAZE CUP • PRINTING/BUILD • REUSABLE • CHECKED 2026-05-27

NEED 6 boards • SPARE 1 spare • BUY 1 • ON HAND no • LOCAL BETTER yes

PICK local print and plastic sleeves • FIND local purchase • CHEAPER Laminate paper mazes

PURPOSE Maze simulation

SAFETY No magnets in pockets.

STORAGE Magnetic maze bin

SOURCE <https://www.sciencemuseumgroup.org.uk/learning/resources/magnetic-maze>

NOTES Table-based.

## Cornstarch and zip bags

REQUIRED

PY-STEM • PYS-02 • OOBLECK ARMOR ARENA • GROCERY • CONSUMABLE • CHECKED 2026-05-27

NEED 6 team stations • SPARE 15 percent extra • BUY 1 • ON HAND no • LOCAL BETTER yes

PICK cornstarch bulk and zip bags • FIND Amazon search: cornstarch bulk zip bags • CHEAPER Local grocery

PURPOSE Non-Newtonian armor pads

SAFETY Corn allergy note. No eating. Double-bag.

STORAGE Oobleck bin

SOURCE <https://www.sciencebuddies.org/stem-activities/oobleck-colloid>

NOTES Clean immediately.

## Cardboard, skewers, straws, craft sticks

REQUIRED

PY-STEM • PYS-03 • CARDBOARD AUTOMATA ARCADE • BUILD SUPPLIES • CONSUMABLE • CHECKED 2026-05-27

NEED 6 team kits • SPARE 15 percent extra • BUY 1 • ON HAND maybe yes • LOCAL BETTER yes

PICK craft sticks skewers straws bulk • FIND Amazon search: craft sticks skewers straws bulk • CHEAPER Reuse cardboard

PURPOSE Cam and follower automata

SAFETY Hot glue staff-supervised.

STORAGE Automata bin

SOURCE <https://www.exploratorium.edu/tinkering/projects/cardboard-automata>

NOTES Prep sample cam.

## Plastic funnels and flexible tubing

RECOMMENDED

UNVERIFIED

PY-STEM • PYS-04 • STETHOSCOPE SPRINT AND RECOVERY CHALLENGE • TUBING/FUNNELS • REUSABLE • CHECKED 2026-05-27

NEED 6 to 12 funnels, 30 ft tubing • SPARE 15 percent extra • BUY 4 • ON HAND maybe no • LOCAL BETTER yes

PICK Norpro Plastic Funnel Set of 3 and vinyl

tubing • FIND [amazon.com/dp/B000HJBFC6](https://www.amazon.com/dp/B000HJBFC6) • ASIN B000HJBFC6 • CHEAPER Dollar-store funnels and hardware-store tubing

PURPOSE Homemade stethoscope

SAFETY Sanitize shared parts. No forced heart-rate measurement.

STORAGE Biomedical bin

SOURCE <https://www.sciencebuddies.org/stem-activities/make-stethoscope>

NOTES Price hidden or not confirmed.



## Sanitizing wipes

REQUIRED

PY-STEM · PYS-04 · STETHOSCOPE SPRINT AND RECOVERY CHALLENGE · SANITATION · CONSUMABLE · CHECKED 2026-05-27

**NEED** 1 tub · **SPARE** 1 spare if available · **BUY** 1 · **ON HAND** maybe yes · **LOCAL BETTER** yes

**PICK** disinfecting wipes classroom · **FIND** Amazon search: disinfecting wipes classroom · **CHEAPER** School nurse supply

**PURPOSE** Shared device cleaning

**SAFETY** Follow site policy.

**STORAGE** Safety bin

**SOURCE** <https://www.sciencebuddies.org/stem-activities/make-stethoscope>

**NOTES** Safety-critical.

## Meter sticks, cones, score sheets

RECOMMENDED

PY-STEM · PYS-05 · REACTION TIME COMBINE · GYM SUPPLIES · REUSABLE · CHECKED 2026-05-27

**NEED** 6 stations · **SPARE** 1 spare meter stick · **BUY** 1 · **ON HAND** maybe yes · **LOCAL BETTER** yes

**PICK** meter sticks and gym cones · **FIND** Amazon search: meter sticks gym cones · **CHEAPER** Borrow from PE/science

**PURPOSE** Reaction time stations

**SAFETY** No sprint collisions. Clear lanes.

**STORAGE** Gym bin

**SOURCE** <https://www.sciencebuddies.org/stem-activities/think-fast-measure-your-reaction-time>

**NOTES** Can be fully seated.

## Original Slinky Jr or metal slinkies

RECOMMENDED

UNVERIFIED

PY-STEM · PYS-06 · SONAR SLINKY SHOWDOWN · WAVES TOOL · REUSABLE · CHECKED 2026-05-27

**NEED** 4 to 6 slinkies · **SPARE** 1 spare · **BUY** 6 · **ON HAND** maybe no · **LOCAL BETTER** yes

**PICK** Original Slinky Jr · **FIND** [amazon.com/dp/B000000IZGI](https://amazon.com/dp/B000000IZGI) · **ASIN** B000000IZGI · **CHEAPER** Borrow from toy/science closet

**PURPOSE** Longitudinal wave modeling

**SAFETY** Finger pinch warning. Do not overstretch.

**STORAGE** Wave bin

**SOURCE** <https://classroom.iihs.org/sonar-slinky/>

**NOTES** Price hidden or sign-in gated.

## Cardstock, foil, tape, paper screens

REQUIRED

PY-STEM · PYS-07 · PINHOLE PRECISION CHALLENGE · OPTICS BUILD · CONSUMABLE · CHECKED 2026-05-27

**NEED** 6 team kits · **SPARE** 15 percent extra · **BUY** 1 · **ON HAND** maybe yes · **LOCAL BETTER** yes

**PICK** aluminum foil cardstock tape · **FIND** Amazon search: aluminum foil cardstock tape · **CHEAPER** Local grocery and school supply

**PURPOSE** Pinhole image viewer

**SAFETY** No direct sun viewing. Pins controlled by staff.

**STORAGE** Optics bin

**SOURCE** <https://www.exploratorium.edu/snacks/personal-pinhole-theater>

**NOTES** Indoor light source works.

## Force-map cards and clipboards

**REQUIRED**

\$10 UNIT

\$10 TOTAL

LOCAL ESTIMATE

PY-STEM · PYS-08 · LOW-ROPES FORCE MAP RELAY · PRINTING · REUSABLE · CHECKED 2026-05-27

**NEED** 6 team sets · **SPARE** 1 spare · **BUY** 1 · **ON HAND** no · **LOCAL BETTER** yes**PICK** local print and laminate · **FIND** local purchase · **CHEAPER** Black and white cards**PURPOSE** Ropes-course physics wraparound**SAFETY** Host ropes rules override module.**STORAGE** Ambler trip folder**SOURCE** [https://www.teachengineering.org/activities/view/wpi\\_empathy\\_activity3](https://www.teachengineering.org/activities/view/wpi_empathy_activity3)**NOTES** Coordinate with Holly.

## Balloons, pop-top caps, recycled CDs

**REQUIRED**

PY-STEM · PYS-09 · HOVERCRAFT HOCKEY HACKATHON · BUILD SUPPLIES · MIXED · CHECKED 2026-05-27

**NEED** 12 to 18 hovercraft kits · **SPARE** 15 percent extra · **BUY** 1 · **ON HAND** maybe no · **LOCAL BETTER** yes**PICK** latex balloons pop top bottle caps · **FIND** Amazon search: latex balloons pop top bottle caps · **CHEAPER** Local party store, collect caps/CDs**PURPOSE** Mini hovercraft build**SAFETY** Latex allergy note. No face-level launches.**STORAGE** Hovercraft bin**SOURCE** <https://www.jpl.nasa.gov/edu/resources/lesson-plan/hovering-on-a-cushion-of-air/>**NOTES** Use nonlatex fallback if needed.

## LED and safe classroom light sources

**RECOMMENDED**

PY-STEM · PYS-10 · SPECTRA SLEUTH SHOWDOWN · LIGHT SOURCES · REUSABLE · CHECKED 2026-05-27

**NEED** 6 stations · **SPARE** 1 spare · **BUY** 1 · **ON HAND** maybe yes · **LOCAL BETTER** yes**PICK** LED flashlight color light set · **FIND** Amazon search: led flashlight color light set · **CHEAPER** Borrow lamps/LEDs**PURPOSE** Mystery spectra stations**SAFETY** No lasers. Avoid hot bulbs.**STORAGE** Spectra bin**SOURCE** <https://www.sciencehistory.org/visit/exhibitions/history-of-fireworks/>**NOTES** Use safe lights.

## Bin cards, address labels, order deck, route mat

**REQUIRED**

\$20 UNIT

\$20 TOTAL

LOCAL ESTIMATE

PY-STEM · PYS-11 · BOOKBOT BIN LOGIC CHALLENGE · PRINTING/BUILD · REUSABLE · CHECKED 2026-05-27

**NEED** 6 sets · **SPARE** 1 spare · **BUY** 1 · **ON HAND** no · **LOCAL BETTER** yes**PICK** local print and laminate · **FIND** local purchase · **CHEAPER** Index cards and painter tape**PURPOSE** ASRS addressing simulation**SAFETY** Keep clear of actual library operations.**STORAGE** BookBot folder**SOURCE** <https://library.temple.edu/spaces/charles-in-charge>**NOTES** Frame as bin addressing and search, not throughput.

## Foam board, dowels, binder clips, toy cart

REQUIRED

PY-STEM · PYS-12 · ACCESSIBILITY RAMP RESCUE LAB · BUILD SUPPLIES · MIXED · CHECKED 2026-05-27

**NEED** 6 team kits · **SPARE** 15 percent extra · **BUY** 1 · **ON HAND** maybe no · **LOCAL BETTER** yes

**PICK** foam board dowels binder clips toy cart · **FIND** Amazon search: foam board dowels binder clips toy cart · **CHEAPER** Borrow toy cars, use cardboard

**PURPOSE** Portable ramp model

**SAFETY** Keep loads low. No standing on models.

**STORAGE** Accessibility bin

**SOURCE** [https://www.teachengineering.org/activities/view/wpi\\_empathy\\_activity3](https://www.teachengineering.org/activities/view/wpi_empathy_activity3)

**NOTES** ADA-style slope model.

## Safety goggles, class set

REQUIRED

SHARED · ALL · ALL ACTIVITIES · PPE · REUSABLE · CHECKED 2026-05-27

**NEED** 30 goggles · **SPARE** 0 if existing · **BUY** 1 · **ON HAND** maybe yes · **LOCAL BETTER** yes

**PICK** classroom safety goggles 30 pack · **FIND** Amazon search: classroom safety goggles 30 pack · **CHEAPER** Borrow from science stockroom

**PURPOSE** Eye protection for splash, soil, plant peel, and oobleck activities

**SAFETY** Sanitize between users.

**STORAGE** PPE bin

**SOURCE** ACS/NSTA general lab safety

**NOTES** Safety-critical.

## Nitrile gloves, 100 pack

REQUIRED

SHARED · ALL · ALL ACTIVITIES · PPE · CONSUMABLE · CHECKED 2026-05-27

**NEED** 2 boxes · **SPARE** 15 percent built into second box · **BUY** 2 · **ON HAND** maybe yes · **LOCAL BETTER** yes

**PICK** nitrile gloves medium 100 pack · **FIND** Amazon search: nitrile gloves medium 100 pack · **CHEAPER** School stockroom

**PURPOSE** Soil, plant, oobleck, and cleanup handling

**SAFETY** Latex-free preferred.

**STORAGE** PPE bin

**SOURCE** ACS/NSTA general lab safety

**NOTES** Safety-critical.

## Clipboards

RECOMMENDED

SHARED · ALL · ALL ACTIVITIES · MEASUREMENT · REUSABLE · CHECKED 2026-05-27

**NEED** 30 · **SPARE** 0 if existing · **BUY** 1 · **ON HAND** maybe yes · **LOCAL BETTER** yes

**PICK** clipboards classroom pack · **FIND** Amazon search: clipboards classroom pack · **CHEAPER** Borrow from department

**PURPOSE** Outdoor and room data collection

**STORAGE** Field bin

**SOURCE** planning line

**NOTES** Reusable.

## Stopwatches or phone timers

RECOMMENDED

SHARED · ALL · ALL ACTIVITIES · MEASUREMENT · REUSABLE · CHECKED 2026-05-27

**NEED** 6 to 12 · **SPARE** 0 if phones allowed · **BUY** 1 · **ON HAND** maybe yes · **LOCAL BETTER** yes

**PICK** digital stopwatch 12 pack · **FIND** Amazon search: digital stopwatch 12 pack · **CHEAPER** Use projected timer plus staff phones

**PURPOSE** Timing races, trials, and relays

**STORAGE** Measurement bin

**SOURCE** planning line

**NOTES** Reusable.

## Masking tape and painter tape

REQUIRED

SHARED · ALL · ALL ACTIVITIES · SUPPLIES · CONSUMABLE · CHECKED 2026-05-27

**NEED** 6 rolls · **SPARE** 1 spare roll · **BUY** 1 · **ON HAND** maybe yes · **LOCAL BETTER** yes

**PICK** masking tape multi pack · **FIND** Amazon search: masking tape multi pack · **CHEAPER** School supply closet

**PURPOSE** Lane marking, labels, and safe floor routes

**SAFETY** Check floor compatibility before use.

**STORAGE** Tape bin

**SOURCE** planning line

**NOTES** Avoid residue.

## Cardstock, copy paper, markers

REQUIRED

SHARED · ALL · ALL ACTIVITIES · SUPPLIES · CONSUMABLE · CHECKED 2026-05-27

**NEED** 500 to 800 pages plus markers · **SPARE** 15 percent print buffer · **BUY** 2 · **ON HAND** maybe yes · **LOCAL BETTER** yes

**PICK** cardstock 250 sheets and markers · **FIND** Amazon search: cardstock 250 pack markers · **CHEAPER** Local print shop or school supply

**PURPOSE** Cards, score slips, station signs, and builds

**STORAGE** Print bin

**SOURCE** planning line

**NOTES** Printing lines separated in Printing sheet.

## Vinyl STEM stickers

PRIZE

SHARED · REWARDS · PRIZE TABLE · PRIZE · CONSUMABLE · CHECKED 2026-05-27

**NEED** 75 to 100 · **SPARE** 15 percent extra · **BUY** 2 · **ON HAND** no · **LOCAL BETTER** yes

**PICK** STEM stickers 100 pack · **FIND** Amazon search: stem stickers 100 pack · **CHEAPER** Local school store

**PURPOSE** 1 to 2 ticket prize

**STORAGE** Prize bin

**SOURCE** planning line

**NOTES** Non-candy.

## Mechanical pencils and gel pens

PRIZE

SHARED · REWARDS · PRIZE TABLE · PRIZE · CONSUMABLE · CHECKED 2026-05-27

**NEED** 50 to 75 · **SPARE** 15 percent extra · **BUY** 2 · **ON HAND** no · **LOCAL BETTER** yes

**PICK** mechanical pencils gel pens bulk · **FIND** Amazon search: mechanical pencils gel pens bulk · **CHEAPER** School supply closet

**PURPOSE** 1 to 4 ticket prize

**STORAGE** Prize bin

**SOURCE** planning line

**NOTES** Useful prize.

## Mini notebooks

PRIZE

SHARED · REWARDS · PRIZE TABLE · PRIZE · CONSUMABLE · CHECKED 2026-05-27

**NEED** 30 · **SPARE** 15 percent extra · **BUY** 1 · **ON HAND** no · **LOCAL BETTER** yes

**PICK** mini notebooks bulk · **FIND** Amazon search: mini notebooks bulk · **CHEAPER** Local dollar store

**PURPOSE** 2 to 4 ticket prize

**STORAGE** Prize bin

**SOURCE** planning line

**NOTES** Non-candy.

## Fidget puzzles or mini desk puzzles

PRIZE

SHARED · REWARDS · PRIZE TABLE · PRIZE · CONSUMABLE · CHECKED 2026-05-27

**NEED** 20 to 30 · **SPARE** 15 percent extra · **BUY** 1 · **ON HAND** no · **LOCAL BETTER** yes

**PICK** mini desk puzzles bulk · **FIND** Amazon search: mini desk puzzles bulk · **CHEAPER** Local store

**PURPOSE** 4 to 7 ticket prize

**SAFETY** Avoid sharp or weapon-like toys.

**STORAGE** Prize bin

**SOURCE** planning line

**NOTES** Middle-school appropriate.

## Safety goggles, class set

REQUIRED

SHARED · ALL · ALL ACTIVITIES · PPE · REUSABLE · CHECKED 2026-05-27

**NEED** 30 goggles · **SPARE** 0 if existing · **BUY** 1 · **ON HAND** maybe yes · **LOCAL BETTER** yes

**PICK** classroom safety goggles 30 pack · **FIND** Amazon search: classroom safety goggles 30 pack · **CHEAPER** Borrow from science stockroom

**PURPOSE** Eye protection for splash, soil, plant peel, and oobleck activities

**SAFETY** Sanitize between users.

**STORAGE** PPE bin

**SOURCE** ACS/NSTA general lab safety

**NOTES** Safety-critical.

## Nitrile gloves, 100 pack

REQUIRED

SHARED · ALL · ALL ACTIVITIES · PPE · CONSUMABLE · CHECKED 2026-05-27

NEED 2 boxes · SPARE 15 percent built into second box · BUY 2 · ON HAND maybe yes · LOCAL BETTER yes

PICK nitrile gloves medium 100 pack · FIND Amazon search: nitrile gloves medium 100 pack · CHEAPER School stockroom

PURPOSE Soil, plant, oobleck, and cleanup handling

SAFETY Latex-free preferred.

STORAGE PPE bin

SOURCE ACS/NSTA general lab safety

NOTES Safety-critical.

## First aid kit and incident log binder

REQUIRED

SHARED · ALL · ALL ACTIVITIES · SAFETY · REUSABLE · CHECKED 2026-05-27

NEED 1 · SPARE 0 · BUY 1 · ON HAND maybe yes · LOCAL BETTER yes

PICK first aid kit classroom · FIND Amazon search: classroom first aid kit · CHEAPER School nurse / camp office

PURPOSE Emergency response and documentation

SAFETY Do not replace official site procedures.

STORAGE Safety cabinet

SOURCE planning line

NOTES Safety-critical.

## Sunscreen, water station, weather plan signs

REQUIRED

\$25 UNIT

\$25 TOTAL

LOCAL ESTIMATE

SHARED · ALL · OUTDOOR MODULES · SAFETY · CONSUMABLE · CHECKED 2026-05-27

NEED 1 set · SPARE extra water · BUY 1 · ON HAND maybe yes · LOCAL BETTER yes

PICK local purchase · FIND local purchase · CHEAPER Camp office supply

PURPOSE Outdoor safety support

SAFETY Allergy-sensitive sunscreen policy.

STORAGE Outdoor safety bin

SOURCE planning line

NOTES Confirm site policy.

## Student handout packet printing

REQUIRED

\$75 UNIT

\$75 TOTAL

LOCAL ESTIMATE

SHARED · PRINTING · ALL PACKETS · PRINTING · CONSUMABLE · CHECKED 2026-05-27

NEED 50 packets or as needed · SPARE 15 percent extra · BUY 1 · ON HAND no · LOCAL BETTER yes

PICK local print · FIND local purchase · CHEAPER Black and white with spot color

PURPOSE Camp handout packets and individual PDFs

STORAGE Print folder

SOURCE planning line

NOTES Estimate only.

## Station signs, score slips, tokens, certificates

REQUIRED

\$60 UNIT

\$60 TOTAL

LOCAL ESTIMATE

SHARED · PRINTING · ALL SIGNS · PRINTING · CONSUMABLE · CHECKED 2026-05-27

NEED 500 pages · SPARE 15 percent extra · BUY 1 · ON HAND no · LOCAL BETTER yes

PICK local print · FIND local purchase · CHEAPER Use cardstock only for durable items

PURPOSE Printable competition kit

STORAGE Print folder

SOURCE planning line

NOTES Estimate only.

## trays, soil, string or yarn roots, craft sticks

BACKUP-ONLY

FROM TREES TO TECH · TTB-01 · ROOT GRIP EROSION RESCUE · BACKUP KIT · MIXED · CHECKED 2026-05-27

NEED 6 team kits · SPARE 15 percent extra · BUY 1 · ON HAND maybe yes · LOCAL BETTER yes

PICK Amazon search for listed materials · FIND Amazon search: trays soil string or yarn roots · CHEAPER Borrow or use existing supplies

PURPOSE Backup-only activity materials

SAFETY Use trays and wipe spills.

STORAGE TTB-01 backup bin

SOURCE EPA green infrastructure and erosion lessons

NOTES Backup-only. Do not order unless needed.

## clinometers, string, washers, rulers

BACKUP-ONLY

FROM TREES TO TECH · TTB-02 · TREE HEIGHT TRIANGULATION SHOOTOUT · BACKUP KIT · MIXED · CHECKED 2026-05-27

NEED 6 team kits · SPARE 15 percent extra · BUY 1 · ON HAND maybe yes · LOCAL BETTER yes

PICK Amazon search for listed materials · FIND Amazon search: clinometers string washers · CHEAPER Borrow or use existing supplies

PURPOSE Backup-only activity materials

SAFETY Outdoor boundaries.

STORAGE TTB-02 backup bin

SOURCE GLOBE Trees Activities

NOTES Backup-only. Do not order unless needed.

## thermometers or IR thermometers, maps, clipboards

BACKUP-ONLY

FROM TREES TO TECH · TTB-03 · URBAN HEAT SHADE DASH · BACKUP KIT · MIXED · CHECKED 2026-05-27

NEED 6 team kits · SPARE 15 percent extra · BUY 1 · ON HAND maybe yes · LOCAL BETTER yes

PICK Amazon search for listed materials · FIND Amazon search: thermometers or IR thermometers maps clipboards · CHEAPER Borrow or use existing supplies

PURPOSE Backup-only activity materials

SAFETY Check heat index. Walk, do not run.

STORAGE TTB-03 backup bin

SOURCE NASA MyNASADData Urban Heat Island

NOTES Backup-only. Do not order unless needed.

## spinach leaves, baking soda solution, cups, oral syringes without needles

BACKUP-ONLY

FROM TREES TO TECH · TTB-04 · PHOTOSYNTHESIS FLOAT-OFF PLAYOFFS · BACKUP KIT · MIXED · CHECKED 2026-05-27

**NEED** 6 team kits · **SPARE** 15 percent extra · **BUY** 1 · **ON HAND** maybe yes · **LOCAL BETTER** yes

**PICK** Amazon search for listed materials · **FIND** Amazon search: spinach leaves baking soda solution cups · **CHEAPER** Borrow or use existing supplies

**PURPOSE** Backup-only activity materials

**SAFETY** No tasting. Handle syringes as lab tools.

**STORAGE** TTB-04 backup bin

**SOURCE** Exploratorium Photosynthetic Floatation

**NOTES** Backup-only. Do not order unless needed.

## pencils, foam, cardboard, weights

BACKUP-ONLY

PY-STEM · PYB-01 · PENCIL PRESSURE SAFETY LAB · BACKUP KIT · MIXED · CHECKED 2026-05-27

**NEED** 6 team kits · **SPARE** 15 percent extra · **BUY** 1 · **ON HAND** maybe yes · **LOCAL BETTER** yes

**PICK** Amazon search for listed materials · **FIND** Amazon search: pencils foam cardboard · **CHEAPER** Borrow or use existing supplies

**PURPOSE** Backup-only activity materials

**SAFETY** No skin tests. Use foam only.

**STORAGE** PYB-01 backup bin

**SOURCE** IIHS Pencil Pressure

**NOTES** Backup-only. Do not order unless needed.

## dominoes, rulers, tape, index cards

BACKUP-ONLY

PY-STEM · PYB-02 · DOMINO NEURON RELAY · BACKUP KIT · MIXED · CHECKED 2026-05-27

**NEED** 6 team kits · **SPARE** 15 percent extra · **BUY** 1 · **ON HAND** maybe yes · **LOCAL BETTER** yes

**PICK** Amazon search for listed materials · **FIND** Amazon search: dominoes rulers tape · **CHEAPER** Borrow or use existing supplies

**PURPOSE** Backup-only activity materials

**SAFETY** Low risk.

**STORAGE** PYB-02 backup bin

**SOURCE** Exploratorium Domino Effect

**NOTES** Backup-only. Do not order unless needed.

## classroom pulleys, string, weights, spring scales if available

BACKUP-ONLY

PY-STEM · PYB-03 · PULLEY RESCUE RELAY · BACKUP KIT · MIXED · CHECKED 2026-05-27

**NEED** 6 team kits · **SPARE** 15 percent extra · **BUY** 1 · **ON HAND** maybe yes · **LOCAL BETTER** yes

**PICK** Amazon search for listed materials · **FIND** Amazon search: classroom pulleys string weights · **CHEAPER** Borrow or use existing supplies

**PURPOSE** Backup-only activity materials

**SAFETY** Keep feet clear of loads.

**STORAGE** PYB-03 backup bin

**SOURCE** TryEngineering Pulleys and Force

**NOTES** Backup-only. Do not order unless needed.



## printed barcode cards, dry erase boards, timers

BACKUP-ONLY

PY-STEM · PYB-04 · BARCODE CHECKSUM RESCUE · BACKUP KIT · MIXED · CHECKED 2026-05-27

**NEED** 6 team kits · **SPARE** 15 percent extra · **BUY** 1 · **ON HAND** maybe yes · **LOCAL BETTER** yes

**PICK** Amazon search for listed materials · **FIND** Amazon search: printed barcode cards dry erase boards  
timers · **CHEAPER** Borrow or use existing supplies

**PURPOSE** Backup-only activity materials

**SAFETY** Normal classroom management.

**STORAGE** PYB-04 backup bin

**SOURCE** Computer science unplugged error-detection concepts

**NOTES** Backup-only. Do not order unless needed.

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BUDGET SCENARIOS

3 LINES

| SCENARIO              | FROM<br>TREES TO<br>TECH | PY-<br>STEM  | SHARED<br>SAFETY<br>AND PPE | PRIZES<br>AND<br>REWARDS | PRINTING<br>AND<br>SUPPLIES | BACKUP-<br>ONLY | TOTAL      | ASSUMPTIONS                                                                    |
|-----------------------|--------------------------|--------------|-----------------------------|--------------------------|-----------------------------|-----------------|------------|--------------------------------------------------------------------------------|
| Preferred             | \$604                    | \$311<br>.47 | \$104                       | \$107                    | \$135                       | \$96            | \$1,357.47 | Includes required and recommended purchases with many prices unverified.       |
| Low-cost              | \$438                    | \$177<br>.98 | \$104                       | \$69.55                  | \$101.25                    | \$0             | \$890.78   | Borrow devices, borrow meters, buy only required consumables and safety items. |
| No new specialty kits | \$334                    | \$311<br>.47 | \$104                       | \$107                    | \$135                       | \$0             | \$991.47   | Assumes MudWatt kits already exist or are borrowed.                            |

All amounts in US dollars.

## INVENTORY BINS

32 LINES

| BIN/KIT       | CAMP                     | ACTIVITIES                                           | CONTENTS                                                                                                                                                                     | PACK<br>OWNER | STORAGE<br>LOCATION | RESET NOTES                                                              |
|---------------|--------------------------|------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|---------------------|--------------------------------------------------------------------------|
| TTT-01<br>kit | From<br>Trees<br>to Tech | TTT-01 MudWatt<br>Bioelectric League                 | 6 MudWatt or equivalent microbial fuel<br>cell kits, fresh soil or mud samples,<br>nitrile gloves, labels and masking tape,<br>optional multimeters, handwashing<br>supplies |               |                     | Check<br>consumables,<br>reset score slips,<br>return reusable<br>tools. |
| TTT-02<br>kit | From<br>Trees<br>to Tech | TTT-02 Forest<br>Sensor Sprint                       | 6 clipboards, 6 paper clinometers, 6<br>thermometers or hygrometers, 6 soil<br>moisture meters if available, route cards,<br>pencils                                         |               |                     | Check<br>consumables,<br>reset score slips,<br>return reusable<br>tools. |
| TTT-03<br>kit | From<br>Trees<br>to Tech | TTT-03 Seed<br>Dispersal Derby                       | paper, cardstock, paper clips, masking<br>tape, string, fan or marked test lane                                                                                              |               |                     | Check<br>consumables,<br>reset score slips,<br>return reusable<br>tools. |
| TTT-04<br>kit | From<br>Trees<br>to Tech | TTT-04 Xylem<br>Pipeline Relay                       | paper towels, cotton string, felt strips,<br>cups, food coloring, trays                                                                                                      |               |                     | Check<br>consumables,<br>reset score slips,<br>return reusable<br>tools. |
| TTT-05<br>kit | From<br>Trees<br>to Tech | TTT-05 Greenhouse<br>Climate Controller<br>Challenge | plant profile cards, control dial boards,<br>clipboards, dry erase markers, tour clue<br>sheets                                                                              |               |                     | Check<br>consumables,<br>reset score slips,<br>return reusable<br>tools. |
| TTT-06<br>kit | From<br>Trees<br>to Tech | TTT-06 Pinecone<br>Weather Machine<br>Challenge      | dry pine cones, index cards, wood<br>skewers, spray bottles, rulers, masking<br>tape                                                                                         |               |                     | Check<br>consumables,<br>reset score slips,<br>return reusable<br>tools. |
| TTT-07<br>kit | From<br>Trees<br>to Tech | TTT-07 Pollinator<br>Network Draft and<br>Build      | plant cards, pollinator cards, grid<br>boards, stickers, markers, season tokens                                                                                              |               |                     | Check<br>consumables,<br>reset score slips,<br>return reusable<br>tools. |
| TTT-08<br>kit | From<br>Trees<br>to Tech | TTT-08 Arboretum<br>Eco-Quest                        | clue cards, route map, clipboards,<br>pencils, evidence token bags, optional<br>QR codes                                                                                     |               |                     | Check<br>consumables,<br>reset score slips,<br>return reusable<br>tools. |
| TTT-09<br>kit | From<br>Trees<br>to Tech | TTT-09 Minecraft<br>Tree World<br>Resilience Cup     | Minecraft Education devices and<br>template or paper grid kit, rubric cards,<br>display timer                                                                                |               |                     | Check<br>consumables,<br>reset score slips,<br>return reusable<br>tools. |

| BIN/KIT       | CAMP                     | ACTIVITIES                                          | CONTENTS                                                                                                                 | PACK<br>OWNER | STORAGE<br>LOCATION | RESET NOTES                                                              |
|---------------|--------------------------|-----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|---------------|---------------------|--------------------------------------------------------------------------|
| TTT-10<br>kit | From<br>Trees<br>to Tech | TTT-10 Tree Ring<br>Climate Detective<br>Tournament | printed ring cards, wood cookies or<br>images, answer boards, timers, claim-<br>evidence cards                           |               |                     | Check<br>consumables,<br>reset score slips,<br>return reusable<br>tools. |
| TTT-11<br>kit | From<br>Trees<br>to Tech | TTT-11 Lotus Leaf<br>Surface Sprint                 | wax paper, cardstock, sandpaper strips,<br>masking tape, pipettes, water                                                 |               |                     | Check<br>consumables,<br>reset score slips,<br>return reusable<br>tools. |
| TTT-12<br>kit | From<br>Trees<br>to Tech | TTT-12 Leaf<br>Stomata<br>Microscope<br>Detective   | leaf samples, clear tape, clear nail polish<br>or prepared slides, USB microscopes or<br>magnifiers, grid sheets, gloves |               |                     | Check<br>consumables,<br>reset score slips,<br>return reusable<br>tools. |
| TTB-01<br>kit | From<br>Trees<br>to Tech | TTB-01 Root Grip<br>Erosion Rescue                  | trays, soil, string or yarn roots, craft<br>sticks, watering cups                                                        |               |                     | Check<br>consumables,<br>reset score slips,<br>return reusable<br>tools. |
| TTB-02<br>kit | From<br>Trees<br>to Tech | TTB-02 Tree Height<br>Triangulation<br>Shootout     | clinometers, string, washers, rulers,<br>clipboards                                                                      |               |                     | Check<br>consumables,<br>reset score slips,<br>return reusable<br>tools. |
| TTB-03<br>kit | From<br>Trees<br>to Tech | TTB-03 Urban Heat<br>Shade Dash                     | thermometers or IR thermometers, maps,<br>clipboards                                                                     |               |                     | Check<br>consumables,<br>reset score slips,<br>return reusable<br>tools. |
| TTB-04<br>kit | From<br>Trees<br>to Tech | TTB-04<br>Photosynthesis<br>Float-Off Playoffs      | spinach leaves, baking soda solution,<br>cups, oral syringes without needles,<br>timers                                  |               |                     | Check<br>consumables,<br>reset score slips,<br>return reusable<br>tools. |
| PYS-01<br>kit | PY-<br>STEM              | PYS-01 Magnetic<br>Capsule Maze Cup                 | small neodymium magnets, maze<br>boards, steel tokens or paper clips,<br>timers, barrier sheets                          |               |                     | Check<br>consumables,<br>reset score slips,<br>return reusable<br>tools. |
| PYS-02<br>kit | PY-<br>STEM              | PYS-02 Oobleck<br>Armor Arena                       | cornstarch, water, zip bags, cardboard<br>sleeves, test weights, spoons                                                  |               |                     | Check<br>consumables,<br>reset score slips,<br>return reusable<br>tools. |

| BIN/KIT       | CAMP        | ACTIVITIES                                                | CONTENTS                                                                                                                    | PACK<br>OWNER | STORAGE<br>LOCATION | RESET NOTES                                                              |
|---------------|-------------|-----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|---------------|---------------------|--------------------------------------------------------------------------|
| PYS-03<br>kit | PY-<br>STEM | PYS-03 Cardboard<br>Automata Arcade                       | cardboard, skewers, straws, craft sticks,<br>tape, hot glue station                                                         |               |                     | Check<br>consumables,<br>reset score slips,<br>return reusable<br>tools. |
| PYS-04<br>kit | PY-<br>STEM | PYS-04<br>Stethoscope Sprint<br>and Recovery<br>Challenge | funnels, flexible tubing, tape, balloons or<br>membranes, stopwatches, sanitizing<br>wipes                                  |               |                     | Check<br>consumables,<br>reset score slips,<br>return reusable<br>tools. |
| PYS-05<br>kit | PY-<br>STEM | PYS-05 Reaction<br>Time Combine                           | meter sticks, stopwatches, cones, score<br>sheets, clipboards                                                               |               |                     | Check<br>consumables,<br>reset score slips,<br>return reusable<br>tools. |
| PYS-06<br>kit | PY-<br>STEM | PYS-06 SONAR<br>Slinky Showdown                           | 4 to 6 metal slinkies, floor tape, station<br>cards, clipboards                                                             |               |                     | Check<br>consumables,<br>reset score slips,<br>return reusable<br>tools. |
| PYS-07<br>kit | PY-<br>STEM | PYS-07 Pinhole<br>Precision Challenge                     | cardstock, foil, tape, pushpins or paper<br>clips, white screen paper, light source                                         |               |                     | Check<br>consumables,<br>reset score slips,<br>return reusable<br>tools. |
| PYS-08<br>kit | PY-<br>STEM | PYS-08 Low-Ropes<br>Force Map Relay                       | element cards, clipboards, center-of-<br>mass templates, pencils                                                            |               |                     | Check<br>consumables,<br>reset score slips,<br>return reusable<br>tools. |
| PYS-09<br>kit | PY-<br>STEM | PYS-09 Hovercraft<br>Hockey Hackathon                     | recycled CDs or plastic discs, pop-top<br>bottle caps, balloons, hot glue station,<br>targets, floor tape                   |               |                     | Check<br>consumables,<br>reset score slips,<br>return reusable<br>tools. |
| PYS-10<br>kit | PY-<br>STEM | PYS-10 Spectra<br>Sleuth Showdown                         | diffraction glasses or gratings, LED<br>sources, incandescent source if<br>available, spectrum cards, museum clue<br>sheets |               |                     | Check<br>consumables,<br>reset score slips,<br>return reusable<br>tools. |
| PYS-11<br>kit | PY-<br>STEM | PYS-11 BookBot Bin<br>Logic Challenge                     | bin cards, address labels, order deck,<br>route mat, timers, optional grabber tool                                          |               |                     | Check<br>consumables,<br>reset score slips,<br>return reusable<br>tools. |

| BIN/KIT       | CAMP        | ACTIVITIES                                 | CONTENTS                                                                 | PACK<br>OWNER | STORAGE<br>LOCATION | RESET NOTES                                                              |
|---------------|-------------|--------------------------------------------|--------------------------------------------------------------------------|---------------|---------------------|--------------------------------------------------------------------------|
| PYS-12<br>kit | PY-<br>STEM | PYS-12<br>Accessibility Ramp<br>Rescue Lab | foam board, craft sticks, dowels, binder<br>clips, rulers, weighted cart |               |                     | Check<br>consumables,<br>reset score slips,<br>return reusable<br>tools. |
| PYB-01<br>kit | PY-<br>STEM | PYB-01 Pencil<br>Pressure Safety<br>Lab    | pencils, foam, cardboard, weights,<br>graph paper                        |               |                     | Check<br>consumables,<br>reset score slips,<br>return reusable<br>tools. |
| PYB-02<br>kit | PY-<br>STEM | PYB-02 Domino<br>Neuron Relay              | dominoes, rulers, tape, index cards                                      |               |                     | Check<br>consumables,<br>reset score slips,<br>return reusable<br>tools. |
| PYB-03<br>kit | PY-<br>STEM | PYB-03 Pulley<br>Rescue Relay              | classroom pulleys, string, weights,<br>spring scales if available        |               |                     | Check<br>consumables,<br>reset score slips,<br>return reusable<br>tools. |
| PYB-04<br>kit | PY-<br>STEM | PYB-04 Barcode<br>Checksum Rescue          | printed barcode cards, dry erase<br>boards, timers                       |               |                     | Check<br>consumables,<br>reset score slips,<br>return reusable<br>tools. |

## SOURCE VERIFICATION LOG

9 LINES

| SOURCE OR ITEM                               | URL                                                                                                                                                         | STATUS                       | LAST CHECKED DATE | USED FOR                              | NOTES                                                                 |
|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|-------------------|---------------------------------------|-----------------------------------------------------------------------|
| Science History Institute Flash! Bang! Boom! | <a href="https://www.sciencehistory.org/visit/exhibitions/history-of-fireworks/">https://www.sciencehistory.org/visit/exhibitions/history-of-fireworks/</a> | verified official web source | 2026-05-27        | PYS-10 Spectra Sleuth and trip tie-in | On view through July 31, 2027 per official source.                    |
| Temple Libraries BookBot                     | <a href="https://library.temple.edu/spaces/charles-in-charge">https://library.temple.edu/spaces/charles-in-charge</a>                                       | verified official web source | 2026-05-27        | PYS-11 BookBot Bin Logic Challenge    | ASRS with nearly 1.5 million volumes per official source.             |
| Temple Ambler Greenhouse                     | <a href="https://ambler.temple.edu/campus-resources/greenhouse">https://ambler.temple.edu/campus-resources/greenhouse</a>                                   | verified official web source | 2026-05-27        | TTT-05 Greenhouse Climate Controller  | Environmental controls and weather station.                           |
| micro:bit data logging                       | <a href="https://microbit.org/get-started/user-guide/data-logging/">https://microbit.org/get-started/user-guide/data-logging/</a>                           | verified official web source | 2026-05-27        | Source audit only                     | Not in final procurement because micro:bit activity was not selected. |
| JUNAN magnets B09V14FGQF                     | <a href="https://www.amazon.com/dp/B09V14FGQF">https://www.amazon.com/dp/B09V14FGQF</a>                                                                     | verified Amazon page         | 2026-05-27        | PYS-01                                | Current public price found: 6.99.                                     |
| Boom 6×1.5 magnets B0BV6P6YYS                | <a href="https://www.amazon.com/dp/B0BV6P6YYS">https://www.amazon.com/dp/B0BV6P6YYS</a>                                                                     | verified Amazon page         | 2026-05-27        | PYS-01                                | Current public price found: 19.49.                                    |
| Diffraction glasses B00JV06CV6               | <a href="https://www.amazon.com/dp/B00JV06CV6">https://www.amazon.com/dp/B00JV06CV6</a>                                                                     | unverified price             | 2026-05-27        | PYS-10 and prizes                     | Product page visible, price not reliably exposed.                     |
| Slinky Jr B00000IZGI                         | <a href="https://www.amazon.com/dp/B00000IZGI">https://www.amazon.com/dp/B00000IZGI</a>                                                                     | unverified price             | 2026-05-27        | PYS-06 and prizes                     | Product page visible, price not reliably exposed.                     |
| MudWatt                                      | <a href="https://www.amazon.com/s?k=MudWatt+STEM+kit">https://www.amazon.com/s?k=MudWatt+STEM+kit</a>                                                       | search URL only              | 2026-05-27        | TTT-01                                | Inventory or same-day procurement check required.                     |